

PHASE 1 MASTER DEVELOPMENT PLAN V3

Ultra-Advanced, Zero-Miss, Existing-Code-First Lead-to-Close Execution Blueprint

Production Engineering Contract for Frontend + Backend + Database + APIs + Events + Jobs
+ AI Agents + WhatsApp + CRM + Quotation + Admin Governance + Observability + QA +
Release

Locked scope: Lead -> Close / valid WON only. Five agents: Sales, Scheduler, Quotation Master, Coordination, Orchestrator/Router. Phase 2 starts only after a valid governed handoff at the WON boundary.

Control	Locked rule
Execution	Inspect -> classify -> implement/correct -> test -> verify -> re-test -> record evidence -> proceed
Existing code	Never assume missing; objective evidence is required before marking complete
Authority	Backend + policy + state + audit are authoritative; AI output cannot self-authorize
Blocker	Stop exactly where a human, credential, provider, account, policy, or business decision is required
Delivery	Exact tested release candidate = exact deployed build
Admin	Enterprise control center for Phase 1 only; architecture must accept later modules without rebuild
Prepared	2026-09-11

Source basis: the supplied AgencyOS Phase 1 flow, five agent responsibility documents, responsibility matrix, interaction/handoff specification, technical/API/data contract, implementation/testing/DoD, final gap-closure specification, Admin Panel screen-by-screen blueprint, plus the wider 23-document AgencyOS architecture set. This V3 turns those requirements into an implementation and verification sequence rather than replacing them.

1. Document Control, Source Hierarchy & How Claude Must Use This Plan

This document is an execution master. It is not permission to redesign the product. The repository and source PDFs determine what already exists, what is partial, and what must be implemented or corrected.

Priority	Source	How it governs implementation
1	Locked Phase 1 business flow and five agent responsibility specs	Business ownership, workflow responsibilities, agent boundaries, exact close path
2	Phase 1 interaction/handoff + technical/API/data contracts	Task/result/handoff/event schemas, identifiers, state machine, routing, API and integration rules
3	Implementation/testing/DoD + gap closure	Build order, evidence standard, blockers, testing and release gates
4	Admin Panel blueprint	Enterprise Phase 1 control surface, 34 screens, roles, UI states, audit and governance controls
5	Wider 23-document AgencyOS set	Cross-system consistency and future-phase compatibility; do not activate later-phase scope

1.1 Required Claude startup sequence

[] Read all Phase 1 PDFs in the designated Phase 1 source folder.

[] Read all original AgencyOS source PDFs, including the 23-document set.

[] Read the entire repository before changing architecture.

[] Map frontend, backend, database, auth, admin, CRM, WhatsApp, AI/provider integrations, tests, CI/CD, deployment and configuration.

[] Create a writable repository tracker, for example docs/AGENCYOS_PHASE1_EXECUTION_TRACKER.md, from this plan.

[] For every requirement classify: COMPLETE, PARTIAL, MISSING, BROKEN, MOCKED, NEEDS_REVIEW, or BLOCKED.

[] Attach objective evidence: file path, symbol/route/model, test name, API response, migration, screenshot, log or reproducible command.

Do not mark COMPLETE because a screen exists, code compiles, an endpoint returns 200, or an LLM claims success.

1.2 Stop / resume contract

Blocker class	Examples	Required action
Human/business decision	Pricing policy, discount exception, scope ambiguity, approval	Stop and document exact decision required
Credential/account	WhatsApp provider, calendar, model provider, webhook secret	Stop; state required credential/config without exposing secret
External system	Calendar unavailable, provider outage, app permission	Stop at integration boundary; keep domain state truthful
Unsafe ambiguity	Ambiguous acceptance, stale quote, conflicting requirement version	Do not infer; block or ask for governed resolution
Implementation defect	Test failure, invalid transition, duplicate side effect	Fix before moving to next dependent unit

2. Phase 1 Target Architecture - Lead to Governed WON

Canonical path: external/client/WhatsApp input -> adapter/auth/identity -> event -> domain state -> Coordination task -> Orchestrator routing -> specialist execution -> normalization -> schema/business validation -> evidence -> Coordination state transition -> next task/wait/retry/escalate -> governed close.

Layer	Responsibility	Phase 1 implementation result
Channel adapters	WhatsApp/other configured inbound and outbound transport	Authenticated, normalized, idempotent message events
Domain	CRM lead/client/conversation/requirements/meeting/quote/ap proval/close state	Authoritative persisted business state
Coordination	Durable tasks, dependencies, blockers, handoffs, retries, evidence	Truthful workflow movement
Orchestrator	Capability, permission, policy, model/provider route and safe execution control	Auditable route decisions
Specialists	Sales, Scheduler, Quotation Master	Domain work only within bounded authority
Policy/approval	Backend-enforced limits and human governance	Out-of-policy actions become governed approvals
Audit/observability	Correlation, state history, provider trace, critical actions	Reconstructable execution evidence
Admin Panel	Human visibility and controlled commands	Operations/governance control surface

2.1 Five-agent authority boundaries

Agent	Owns	Must not do
Sales	Customer/commercial conversation, qualification, discovery, objections, next action	Invent price, promise exceptions, self-approve discounts, mark ambiguous acceptance
Scheduler	Scheduling intent, availability, proposals, booking, reminders, completion evidence	Fabricate availability or claim booking before provider verification
Quotation Master	Commercial draft/version, scope, price/discount/tax/payment/timeline within configuration	Silently mutate accepted versions or bypass approval policy
Coordination	Workflow state, tasks, dependencies, handoffs, blockers, retry/escalation, evidence acceptance	Choose AI provider/model, change policy, perform specialist work
Orchestrator/Router	Capability matching, permission/policy checks, agent/model/provider routing, safe retry/fallback	Act as business approver or rewrite business rules

Admin Panel role: human governance and operations surface. It is not a second source of truth. Backend re-validates every controlled command against current state, permissions, policy, version and evidence.

3. PHASE 0 - Repository Discovery, Baseline and Traceability Gate

Phase 0 is mandatory even when the repository already contains substantial functionality. The goal is to prevent duplicate implementation, unsafe rewrites and false completion.

ID	Audit area	Evidence required
P0-01	Repository topology	Repo tree, package manifests, application entrypoints, services, workers, migrations, routes
P0-02	Runtime stack	Frontend/backend/framework versions, database, queues, storage, auth, hosting
P0-03	Domain map	Lead/client/conversation/requirements/meeting/quote/approval/task/event/audit models
P0-04	Agent map	Agent implementations, registries, prompts/policies, tools, orchestrator, provider adapters
P0-05	Integration map	WhatsApp, calendar, model providers, webhooks, email/notifications where Phase 1 relevant
P0-06	Admin map	Routes, screens, components, permissions, API bindings, tables, filters, action handlers
P0-07	Testing map	Unit, integration, contract, E2E, security, fixtures, CI and coverage configuration
P0-08	Deployment map	Build pipeline, environments, migrations, health checks, rollback path
P0-09	Gap classification	Each plan requirement classified with objective evidence
P0-10	Baseline gate	Build/test/runtime baseline captured; known blockers listed

3.1 No-rewrite rule

Do not replace a working implementation solely because another pattern looks cleaner. Refactor only when the existing implementation is objectively incompatible with a locked contract, unsafe, untestable, or blocks a required DoD item. Preserve stable behavior with regression tests.

3.2 Tracker record format

Each item must record: ID | Source section | Repository evidence | Current status | Dependencies | Planned change | Test command | Verification evidence | Reviewer/actor | Date | Blocker (if any) | Re-test result.

4. Foundation - Tenancy, Identity, IDs, State, Audit and Security

4.1 Tenant/auth/RBAC foundation

- [] organization_id is mandatory tenant boundary for Phase 1 resources.
- [] Backend authorization is enforced on every read/write and controlled command.
- [] Roles include configured Super Admin, Admin, Sales, Operations and Auditor boundaries from the Admin blueprint.
- [] Cross-tenant access is rejected and tested.
- [] Secrets are never returned to agents, logs, admin UI or client responses.

4.2 Canonical IDs and trace

ID	Purpose	Rule
organization_id	Tenant scope	Every Phase 1 resource query/write carries or derives tenant scope
client_id	Stable client identity	Separate from lead; safe identity resolution
lead_id	Sales opportunity	Stable across conversation and commercial journey
task_id	Work item	Unique, durable, traceable
event_id	Event identity	Idempotent event processing
correlation_id	End-to-end trace	Preserved across agent/provider/workflow hops
approval_id	Governed decision	Exact action/version reference
quote_id + quote_version	Commercial identity	Material changes produce immutable version
requirement_version	Scope identity	Quote uses explicit confirmed/approved version
provider_object_id	External object	Never replaces AgencyOS internal identity
idempotency_key	Duplicate-side-effect protection	Required on side-effecting operations

4.3 Authoritative state machine

Task lifecycle: CREATED -> VALIDATING -> READY -> DISPATCHED -> ACKNOWLEDGED -> IN_PROGRESS -> WAITING_FOR_RESULT -> RESULT_RECEIVED -> VALIDATING_RESULT -> ACCEPTED -> HANDOFF_READY -> HANDED_OFF -> VERIFIED -> CLOSED. Exceptions: BLOCKED, RETRYING, FAILED, EXPIRED, CANCELLED, ESCALATED.

4.4 Security hard gates

- [] Invalid transitions return a conflict/validation error and leave prior state intact.
- [] State transition actor/source, timestamp and correlation are auditable.
- [] Admin/manual overrides are explicit, authorized and logged.
- [] Provider responses are untrusted until normalized, schema-validated and business-validated.
- [] A model output can recommend an action but cannot grant permission, approve a discount or create a payment fact.

5. Contract Layer - Request, Result, Handoff, Event, Approval and Quote

5.1 Canonical task contract

Required fields include task_id, organization/client/lead/project scope where applicable, source agent, destination agent/capability, task type, workflow state, intent, minimum context references, requirement/version refs, artifact/version refs, acceptance criteria, required output schema, policy/approval references, priority, deadline, dependencies, attempt number, retry budget, previous failure summary, idempotency key and timestamps.

5.2 Canonical result contract

Required fields include task/correlation IDs, agent identity/version, result status, structured result, schema version, evidence/artifact references, warnings, blocker, error class, attempt number, provider/model metadata where available, usage/cost metadata where available, validation status, next action and completion evidence.

5.3 Handoff acceptance contract

- [] Source and destination are explicit.
- [] Current workflow state and next expected state are explicit.
- [] Exact requirement/quote/artifact versions are referenced.
- [] Acceptance criteria and policy/approval refs are included.
- [] Dependencies and prior failures are retained.
- [] Receiving agent validates the handoff before executing.
- [] Coordination updates workflow only after result/evidence validation.

5.4 Internal API surface

Method	Endpoint	Purpose
POST	/api/v1/agent-tasks	Create structured task
GET	/api/v1/agent-tasks/{task_id}	Read task/status
POST	/api/v1/agent-tasks/{task_id}/dispatch	Dispatch after validation
POST	/api/v1/agent-tasks/{task_id}/result	Submit structured result
POST	/api/v1/agent-tasks/{task_id}/retry	Safe retry
POST	/api/v1/agent-tasks/{task_id}/cancel	Permitted cancellation
GET	/api/v1/agent-tasks/{task_id}/trace	Execution/handoff trace
POST	/api/v1/events	Publish internal event
POST	/api/v1/webhooks/{provider}	Authenticated provider event
GET	/api/v1/quotes/{quote_id}	Read quote/current version
POST	/api/v1/quotes	Create authorized draft
POST	/api/v1/approvals	Create governed approval
POST	/api/v1/approvals/{approval_id}/decision	Record authenticated decision

6. Workflow Infrastructure - Events, Queues, Jobs, Idempotency, Retry and Recovery

6.1 Event processing

[] Normalize inbound events into an internal envelope with event_id, timestamp, source, type, scope, provider object ID, correlation ID and idempotency metadata.

[] Authenticate webhooks before processing.

[] Reject malformed, unauthorized, stale or duplicate events without mutating authoritative state.

[] Record processing status and rejection reason.

6.2 Job engine

Capability	Required behavior	Verification
Idempotency	Duplicate jobs/events cannot create duplicate booking/message/quote/approval side effects	Replay same request/event; exactly one accepted side effect
Retry	Bounded attempts + backoff + explicit retryable error classes	Transient failure recovers without duplicate state
Timeout	Durable timeout state; no fake success	Provider hangs; domain remains pending/error
Dead letter	Exhausted failures become inspectable	Task appears in recovery queue with trace
Circuit breaker	Repeated provider failures stop pressure	Provider outage blocks safely and recovers after health restored
Resume	Process restart does not lose workflow	Restart worker during in-flight task; resume from durable state
Stale safety	Superseded versions/dependencies block obsolete work	Old requirement/quote task cannot advance current workflow

6.3 Canonical errors

AUTHENTICATION_ERROR, AUTHORIZATION_ERROR, VALIDATION_ERROR, RATE_LIMITED, TIMEOUT, PROVIDER_UNAVAILABLE, NOT_FOUND, CONFLICT/DUPLICATE, PROVIDER_ERROR, NETWORK_ERROR, POLICY_BLOCKED, UNKNOWN_ERROR. Map provider-specific errors into canonical classes before workflow decisions.

7. Orchestrator / Router - Implementation Sequence

ID	Unit	Done evidence
OR-01	Capability registry	Agent IDs/versions, tasks, tools, permissions, data classes, schemas, availability, constraints
OR-02	Eligibility filter	Task support + permissions + data classification + tools + schema compatibility + health
OR-03	Policy scoring	Quality/reliability/latency/cost/capability according to configured routing policy
OR-04	Execution envelope	Scope, intent, context refs, exact versions, output schema, acceptance criteria, policy, deadlines, retry and idempotency
OR-05	Provider adapter	Provider-neutral interface for auth/model/capability/context/tool/rate-limit/health/privacy metadata
OR-06	Result validation	Schema validation + business rule validation before state update
OR-07	Retry/fallback	Safe bounded retry; fallback only where idempotent and policy-permitted
OR-08	Escalation	Exhausted or unsafe failures become visible escalation, not silent success
OR-09	Trace	Record selected agent/model/provider, policy version, attempts, cost/latency metadata and outcome
OR-10	Negative authority tests	Model cannot approve price, permission, acceptance or payment fact

Routing score from technical source: $ROUTE_SCORE = Quality \times PolicyFit \times Availability \times Reliability \times CapabilityFit / (Cost \times LatencyPenalty)$. Implement as configurable/versioned policy, not hard-coded business logic.

8. Coordination Agent - Durable Workflow Controller

ID	Unit	Done evidence
CO-01	Workflow state	Durable current step and transition validation
CO-02	Task lifecycle	Create -> validate -> ready -> dispatch -> execute -> validate result -> accept/hand-off
CO-03	Dependencies	Explicit prerequisite graph; no dispatch before dependency pass
CO-04	Version safety	Reject stale/obsolete requirement, quote and artifact references
CO-05	Handoff packet	Complete shared envelope with acceptance criteria and evidence
CO-06	Blockers	Typed blocker + owner + age + escalation policy
CO-07	Retry	Coordinate safe retry without changing specialist ownership
CO-08	Recovery	Resume after worker/process interruption
CO-09	Acceptance	Result received is not acceptance; evidence/business validation required
CO-10	Observability	Trace all tasks, transitions, dependencies and handoffs

Coordination must never select an AI provider/model, change business policy, approve commercial exceptions, perform specialist work, or advance a gated state without evidence.

9. Sales Agent + CRM - Lead-to-Requirements Execution

9.1 Lead lifecycle

Required source states: NEW, CONTACTED, RESPONDING, QUALIFIED, REQUIREMENTS_PENDING, QUOTE_DRAFT, QUOTE_SENT, NEGOTIATION, AWAITING_ACCEPTANCE, PAYMENT_PENDING, WON, LOST, NURTURE, DISQUALIFIED.

9.2 Implementation units

ID	Unit	Done evidence
SA-01	Lead intake + normalization	Source mapped; contact data normalized; tenant scoped
SA-02	Dedupe + identity	Phone/email/client/active/recent-lost lead matching; uncertain matches require review
SA-03	Conversation context	Structured CRM context loaded; WhatsApp history is evidence, not source of truth
SA-04	Qualification	Service, users, platform, features, integrations, design, timeline, budget, urgency, decision-maker, assets, language, payment expectations
SA-05	Requirement discovery	Confirmed facts vs assumptions vs unanswered questions; exclusions and optional items
SA-06	Trust + objections	Structured objection categories and approved response/playbook boundaries
SA-07	Quote readiness	Quotation task created only when requirement set meets readiness rules
SA-08	Negotiation	Requested changes classified and routed under policy
SA-09	Acceptance	Only exact quote version accepted by unambiguous authorized evidence
SA-10	Close gate	Payment/approved exception gate verified; then WON
SA-11	Handoff	Structured WON handoff packet ready for Phase 2; no downstream PM activation in Phase 1

10. WhatsApp / Communication Layer

10.1 Channel rules

- [] Inbound webhook signature/authentication is verified before processing.
- [] Message/provider IDs map to internal conversation/message identities with idempotency.
- [] Lead/client mapping occurs through CRM identity resolution.
- [] Outbound messages are rendered from structured data and tracked by provider reference.
- [] Communication state is separate from authoritative CRM/commercial state.
- [] Internal analysis/memory is visibly separated from client-facing content.
- [] Do-not-contact, pause automation and escalation controls are permissioned and audited.

10.2 Required tests

- [] Duplicate inbound message replay.
- [] Out-of-order event.
- [] Unknown contact.
- [] Known client + new opportunity.
- [] Provider timeout on outbound.
- [] Provider rate limit.
- [] Unauthorized webhook.
- [] Duplicate outbound retry.
- [] Human takeover / automation pause where configured.

11. Scheduler Agent - Meeting Lifecycle

Canonical scheduling path: Lead message -> scheduling intent -> mode -> date/time -> timezone -> availability -> slot proposal -> client confirmation -> booking -> admin notification -> reminders -> call/meeting -> completion -> evidence -> analysis -> context update -> Sales trigger.

ID	Unit	Done evidence
SC-01	Intent	Call/meeting/reschedule/cancel mapped
SC-02	Identity	Correct lead/client/opportunity
SC-03	Mode	CALL/VIDEO/IN_PERSON configured
SC-04	Date/time	Explicit/relative/range/daypart resolved into structured candidate
SC-05	Timezone	Stored reliable timezone; canonical timestamps
SC-06	Availability	Authoritative calendar/resource/working hours/blocks/buffer/min notice/capacity
SC-07	Proposal	Slot + timezone + duration + mode clear
SC-08	Confirmation	Unambiguous client selection; ambiguous "okay" does not book
SC-09	Booking	Recheck availability; provider event + idempotency; verify response
SC-10	Client confirmation	Only after actual provider success
SC-11	Reminder jobs	Timezone-aware, policy-driven, stale-safe after change
SC-12	Reschedule/cancel	History preserved; reminders updated safely
SC-13	Completion	Explicit authorized completion/no-show outcome
SC-14	Evidence	Audio/video/notes/PDF/etc. with permission boundaries
SC-15	Analysis	Grounded structured outcome with provenance
SC-16	Sales reactivation	Context update + next action
SC-17	Failure	Timeout/credentials/outage leave pending/error state

12. Quotation Master - Commercial Version Control

12.1 Commercial input gate

- [] Confirmed/approved requirement_version is present.
- [] Category, complexity, platform, features, integrations, design and timeline inputs are structured.
- [] Pricing, discount, payment, milestone and tax rules are configured and versioned.
- [] Client data, validity, terms and approval requirements are explicit.

12.2 Build sequence

ID	Unit	Done evidence
QM-01	Deterministic calculation	Stored line-item inputs + formula/policy version; reproducible total
QM-02	Discount controls	Bounds + exception triggers
QM-03	Tax/GST	Configured source; never invented
QM-04	Payment schedule	Milestones/payment terms separated from delivery completion state
QM-05	Draft	Quote summary, scope, deliverables, timeline, price, discount, tax, schedule, validity, terms, next action
QM-06	Versioning	Material change = new immutable version; one current/active version
QM-07	Approval	Exact quote/version + policy + actor/role + expiry
QM-08	Delivery	Exact version linked to outbound delivery evidence
QM-09	Negotiation/revision	Every material scope/commercial change creates next version
QM-10	Acceptance	Exact version + identity + timestamp + terms + evidence
QM-11	Post-acceptance lock	Accepted baseline cannot be silently modified

12.3 Approval state

DRAFT -> SUBMITTED -> UNDER_REVIEW -> APPROVED / REJECTED / CHANGES_REQUESTED / EXPIRED / CANCELLED -> EXECUTED -> VERIFIED. Approval does not itself equal execution or acceptance.

13. Admin Approval -> Negotiation -> Exact Acceptance -> WON

13.1 Approval enforcement

[] Out-of-policy discount, special payment/no-advance, special offer, high-value thresholds, missing required inputs and policy conflicts generate approval requirements.

[] WhatsApp/admin approval response is treated as a structured decision only after backend validation of actor, exact version, expiry and policy.

[] Stale/expired/replaced approval cannot execute.

[] Ambiguous response never executes or closes.

13.2 Exact acceptance protocol

Acceptance requires an explicit reference to the exact quote version being accepted. Store identity/actor, timestamp, channel/evidence, accepted version, acceptance terms, and correlation trace. Never infer acceptance from generic replies such as "okay" when the referenced version is unclear.

13.3 Payment/exception gate

Payment-sensitive transition remains separate from quotation acceptance. The system verifies the configured payment or approved exception evidence before allowing WON. Finance is not activated as a Phase 1 agent workforce; Phase 1 only preserves the gate/reference needed to close truthfully.

13.4 WON packet

Field	Required content
Lead/client	Stable IDs + tenant scope + owner
Requirements	Exact approved requirement_version
Commercial	Exact accepted quote_id + quote_version
Approval	Required approval_id(s) and decision evidence
Acceptance	Authorized actor + timestamp + evidence
Payment/exception	Configured verification reference
Workflow	Final state transition + task/handoff trace
Next phase	Structured handoff packet ready; Phase 2 not activated
Audit	Correlation trail across all important events

14. Enterprise Admin Panel - Phase 1 Screen-by-Screen Execution

The blueprint contains 34 screens. Build them inside a scalable shell, but activate only Phase 1 functionality. Every screen must support loading, empty, error, blocked and permission-denied states. No optimistic UI for governed transitions.

ID	Screen	Priority	Execution focus
A01	Admin Overview Dashboard	P0	Lead-to-close metrics, approvals, follow-ups, meetings, quotes, blockers, AI/provider health, critical audit
A02	Global Search	P0	Authorized search by IDs, entities, lead/client/quote/meeting/task/correlation
A03	Lead List	P0	Server filters, saved views, assignment, controlled bulk actions
A04	Lead 360	P0	Overview + conversation + requirements + meetings + quotes + negotiation + tasks + audit
A05	WhatsApp Conversation	P0	Messages, evidence, context, quote links, automation pause/DNC/escalation
A06	Requirements Workspace	P0	Facts/assumptions/questions/version/confirmation/quote readiness
A07	Requirement Compare	P1	Side-by-side revision diff + evidence + quote impact
A08	Scheduler Calendar	P0	Views, filters, timezone, conflict indicators, reschedule/cancel
A09	Meeting Detail	P0	Provider event, reminders, completion, evidence, analysis
A10	Quotation List	P0	Current version, state, approval, validity, stale indicators
A11	Quotation Detail	P0	Exact version, scope, line items, tax, payments, approvals, delivery, acceptance, audit
A12	Quotation Compare	P0	Commercial and scope diff by version
A13	Approval Center	P0	Governance queue, exact action/version, value/risk/expiry
A14	Approval Detail	P0	Decision record + role + evidence + audit
A15	Negotiation Workspace	P1	Objection, rounds, requested changes, policy limits, dependencies
A16	Workflow Task Board	P0	Task states, filters, stale tasks, authorized controls
A17	Task Detail / Trace	P0	Envelope, route, attempts, result, evidence, transitions
A18	Dependency / Blocker	P1	Graph/list of prerequisites, owner, age, escalation
A19	AI Workforce Dashboard	P1	Agent workload/health and operational visibility
A20	Agent Detail	P1	Capability, permissions, tasks, failures, version
A21	Routing / Model Trace	P1	Route candidates, selected route, policy, provider/model trace
A22	Integrations Hub	P0	Provider inventory, status, enable/disable
A23	Integration Detail / Health	P1	Health, limits, errors, test connection where supported
A24	Policies & Controls	P0	Policy visibility and governed configuration
A25	Pricing / Discount Rules	P0	Configured price/discount policies and exception thresholds
A26	Notification Rules	P1	Operational notifications/reminders
A27	Audit Timeline	P0	Critical trace and changes
A28	System Health / Jobs	P0	Queues/jobs/retries/dead letters/provider health
A29	Incidents / Recovery	P0	Failures, escalations, retry/recover actions
A30	Users / Roles / Access	P0	RBAC and access review
A31	Organization Settings	P1	Org-level settings required by Phase 1
A32	Profile / Preferences	P2	User preferences
A33	Readiness / Release	P1	Hard gates, evidence, release candidate state
A34	Help / Documentation	P2	System docs and operator guidance

15. Admin Command Pattern - No UI-Only Authority

Every action button must call a backend command. The backend performs the final authorization, current-state, policy, version, evidence and concurrency checks. On success, authoritative state changes first; the UI refreshes from returned state/evidence.

Action	Frontend	Backend must verify	Evidence
Assign/reassign lead	Command + confirm	RBAC, tenant, current owner/state	actor + timestamp + prior/new owner
Schedule/reschedule/cancel	Structured dialog	RBAC, current meeting state, availability	provider event + internal meeting history
Generate/revise quote	Exact requirement version	Readiness, current version, pricing/policy inputs	quote/version + calculation inputs
Approve/reject quote	Exact approval detail	Role, policy, exact version, expiry, concurrency	approval_id + decision + audit
Retry task	Retry control	Retry budget, error class, idempotency	attempt + failure context
Pause/resume automation	Controlled command	RBAC + state + reason	audit + resulting status
Manual override	High-risk confirmation	Explicit permission + policy + reason	override audit + old/new state

15.1 UI quality gate

- [] Premium enterprise information hierarchy with clean dense layouts.
- [] Semantic text + badges, never color-only status.
- [] Consistent tables, filters, drawers, timelines, confirmation and error patterns.
- [] Keyboard accessibility and visible focus.
- [] Responsive desktop-first layout.
- [] Critical state transitions show exact target state/version before confirmation.

16. AI Memory, Context, Evidence and Provenance

- [] Agent context is minimum necessary, permission-aware and provenance-preserving.
- [] Structured CRM/requirements/quote records are authoritative over raw conversational memory.
- [] Context references point to exact versions rather than copying mutable data blindly.
- [] Stale/conflicting versions are detected before execution.
- [] Sensitive data is excluded unless explicitly required and authorized.
- [] Call/meeting analysis remains evidence-backed and does not silently convert extraction into client-confirmed fact.
- [] All material AI outputs have agent/version, model/provider where available, task/correlation, validation status and evidence references.

16.1 Evidence ladder

Evidence class	Examples	Can advance workflow?
Observation	Model text, heuristic, extraction	Only as input to validation; never as authority
Structured record	Persisted CRM/meeting/requirement value	Yes where state rules permit
External provider proof	Verified booking/event/provider object	Yes for integration-dependent transitions
Human approval	Authenticated approved decision	Yes for governed action
Exact acceptance	Authorized exact quote version acceptance	Yes for acceptance gate
Payment/exception verification	Configured authoritative verification reference	Yes for WON gate where required

17. Observability, Audit and Operational Recovery

17.1 Minimum trace

A complete trace must connect request_id/correlation_id -> event -> domain entity -> task -> agent -> route -> provider/model -> result -> validation -> evidence -> state transition -> next task/handoff. Sensitive payloads are masked.

17.2 Operational dashboards

- [] Task backlog by agent/state/age/priority.
- [] Blocked and escalated workflow items with owner and age.
- [] Retry volume, dead-letter count and oldest failed item.
- [] Provider health, rate limits, timeout/error rates and circuit state.
- [] Quote approval latency, stale approvals and expiring quotes.
- [] Meeting booking failures, no-shows and reminder failures.
- [] Unauthorized/denied actions and security anomalies.
- [] Critical audit changes and admin overrides.

17.3 Recovery rules

Recovery actions must preserve domain truth. Never recover by directly editing database state around business rules. Prefer replaying a canonical command/event or using an authorized administrative recovery command that creates a complete audit trail.

18. 360-Degree Verification Plan

18.1 Test layers

Layer	Coverage
Unit	Domain calculations, state transitions, parsers, validators, policy checks, version comparisons
Contract	Request/result/handoff/event schemas and canonical errors
Integration	DB, queue, provider adapters, WhatsApp, calendar, AI providers, approval path
API	Auth, RBAC, tenant scope, validation, idempotency, concurrency, error mapping
UI	All 34 screens: loading/empty/error/blocked/denied + action/permission behavior
E2E	Full Lead-to-WON flows including branch and failure paths
Security	Tenant isolation, RBAC, secret handling, injection, rate limits, webhook auth
Performance	Queue throughput, latency, concurrency and large-list search/filter
Deployment	Migration, health, startup, smoke, rollback/recovery

18.2 Mandatory functional scenarios

- [] F-01 WhatsApp lead intake + dedupe
- [] F-02 Qualification capture
- [] F-03 Requirements discovery and confirmation
- [] F-04 Meeting intent + slot proposal
- [] F-05 Successful booking with provider reference
- [] F-06 Booking retry without duplicate event
- [] F-07 Quote generated from exact requirement version
- [] F-08 Approval request and decision
- [] F-09 Quote delivery and response tracking
- [] F-10 Negotiation + immutable new quote version
- [] F-11 Ambiguous acceptance blocked
- [] F-12 Exact-version acceptance succeeds
- [] F-13 Payment/approved-exception gate
- [] F-14 WON + Phase 2 handoff packet
- [] F-15 Provider failure leaves truthful pending/error state

19. Negative, Failure, Security and False-Success Test Matrix

Scenario	Expected result	Hard-fail if
Duplicate webhook/event	Idempotent ignore/update with trace	Duplicate side effect occurs
Stale requirement version	Task blocked/rejected	Obsolete scope reaches quote
Stale quote approval	Approval unusable	Older approval executes current quote
Ambiguous "okay"	Clarification required	System marks accepted
Provider timeout	Pending/error + retry/recovery	Success shown without verified external state
Unauthorized admin command	403/authorization failure + audit	State changes
Cross-tenant resource ID	Not found/denied without leakage	Other tenant data exposed
Model says "approved"	Ignored as authority	Commercial action executes
Worker crash during booking	Resume safely/idempotently	Second booking created
Policy-blocked discount	Approval required	Discount silently applied
Accepted quote mutation	Rejected/new version required	Accepted baseline changes silently
Skipped critical test	Release blocked	Readiness still marked green
Unverified deployment	Release blocked	Different build deployed than tested RC

20. Release Candidate, Deployment and Production Gates

20.1 Hard gates

- ☐ Phase 0 repository audit complete and tracker current.
- ☐ No P0 functional requirement remains MISSING/BROKEN/UNVERIFIED.
- ☐ Critical security/tenant/authority tests pass.
- ☐ All mandatory E2E cases pass.
- ☐ No critical hard gate is hidden by aggregate readiness score.
- ☐ Database migrations are verified and reversible where applicable.
- ☐ Provider integrations have configured, tested credentials or are explicitly documented as BLOCKED.
- ☐ Operational health checks, queues, retries and recovery paths are verified.
- ☐ Admin Panel Phase 1 P0 screens are verified end-to-end against backend commands.
- ☐ Release candidate identifier is immutable and traceable to the exact tested source/build.

20.2 Deployment smoke

Deploy the exact release candidate used for acceptance. Verify startup, authentication, tenant isolation, health endpoints, critical APIs, event ingestion, worker execution, Admin login and one representative Lead-to-WON smoke path in a controlled environment. Record evidence.

20.3 Rollback readiness

A rollback plan must identify the deployable artifact, database compatibility constraints, migration strategy, queue/event behavior, provider side effects and operator actions. Rollback cannot claim to reverse an irreversible external side effect; such cases require explicit compensation/reconciliation procedures.

21. Master Execution Checklist - Dependency-Ordered

Master ID	Stage	Coverage
M01	Repository + baseline	P0-01..P0-10
M02	Tenant/auth/RBAC + IDs	Foundation sections
M03	Database/domain/state/audit	Domain + state machine
M04	Contracts	Task/result/handoff/event/approval/quote
M05	Events/jobs/idempotency/retry	Workflow infrastructure
M06	Provider adapters + webhook security	Integration contract
M07	Orchestrator/Router	OR-01..OR-10
M08	Coordination	CO-01..CO-10
M09	Sales + CRM + requirements	SA-01..SA-11
M10	WhatsApp	Channel tests
M11	Scheduler	SC-01..SC-17
M12	Quotation Master	QM-01..QM-11
M13	Approval + negotiation + exact acceptance + WON	Close gates
M14	Admin Panel P0/P1	A01..A34
M15	Observability + incidents/recovery	Trace + dashboards
M16	360 QA + security + E2E	F-01..F-15 + negative matrix
M17	Release candidate + smoke + rollback	Release gates

Sequential rule: do not start a stage whose prerequisites are incomplete or blocked. Independent work may run in parallel only when it cannot mutate or invalidate a pending dependency; the tracker must record the reason for any parallelization.

22. Claude Code Execution Protocol - Exact Working Contract

22.1 Loop for every checklist item

1) READ the relevant source requirements and repository code. 2) INSPECT dependencies and current behavior. 3) CLASSIFY existing implementation. 4) IMPLEMENT or CORRECT only the scoped unit. 5) TEST with automated and targeted checks. 6) VERIFY against the exact acceptance contract. 7) RE-TEST affected regression paths. 8) RECORD objective evidence. 9) MARK the tracker item. 10) PROCEED only when all prerequisites are green.

22.2 Claude is forbidden to

- ☐ Skip repository inspection and rebuild from scratch.
- ☐ Treat a mock/stub/placeholder as production completion.
- ☐ Advance workflow state directly in UI code.
- ☐ Let an LLM output grant permission or business authority.
- ☐ Invent pricing, tax, availability, payment confirmation or requirements.
- ☐ Use raw provider credentials in agent logic.
- ☐ Silently mutate historical quote/requirement versions.
- ☐ Ignore failed tests because the feature appears to work manually.
- ☐ Continue through an explicit human/credential/provider/business-decision blocker.

22.3 Evidence quality rule

For every completed item, evidence should answer: what changed or was verified, where it exists, how it was tested, what exact output/state was observed, and what source requirement it satisfies. Screenshots alone are insufficient for backend/state claims; tests/logs/API traces are required where relevant.

23. Granular Task Definition Template - Use for Every Implementation Unit

Field	Required content
Task ID	Unique ID tied to this plan
Source requirement	PDF + section/contract reference
Objective	One measurable implementation outcome
Current state	COMPLETE/PARTIAL/MISSING/BROKEN/MOCKED/NEEDS_REVIEW/BLOCKED
Repository evidence	Paths, symbols, migrations, routes, tests
Dependencies	Upstream task IDs / external prerequisites
Data contract	Inputs, outputs, IDs, versions
State impact	Allowed before/after transitions
Authority	Agent/policy/admin/backend boundary
UI/API	Screen, endpoint, event, command where applicable
Failure path	Timeout, duplicate, stale, unauthorized, provider failure
Tests	Unit/contract/integration/E2E/security as applicable
Evidence	Command output, API trace, DB state, screenshot, logs
DoD	Exact acceptance criteria
Blocker	Exact required input and stop point
Re-test	Regression result after implementation

23.1 Example - booking

Field	Example
Task ID	SC-09
Objective	Create verified idempotent calendar booking
Dependencies	SC-05, SC-06, provider auth
Checks	recheck availability; idempotency; provider response; internal record
Failure	timeout => pending/error; retry bounded
Evidence	provider event ID/link + internal meeting record + trace
Hard stop	missing calendar credential or authoritative availability

24. Cross-Agent Interaction Matrix

From	To	Trigger	Required handoff	Return / next step
Channel/CRM	Sales	New or updated lead message	Lead/client IDs + message evidence + current state	Qualification/discovery task
Sales	Scheduler	Call/meeting intent	Lead IDs + intent + context + timezone candidates	Proposed/confirmed meeting result
Scheduler	Coordination	Booking/completion event	Meeting ID + provider evidence + result status	Sales reactivation or wait
Sales/Coordination	Quotation Master	Requirements ready	Exact requirement_version + quote intent + policy refs	Quote draft/version
Quotation Master	Policy/Admin	Out-of-policy commercial condition	Exact quote/version + action + evidence	Approval decision
Quotation Master	Sales	Quote delivered/revised	Exact quote/version + delivery evidence	Negotiation/acceptance follow
Sales	Coordination	Acceptance/close signal	Exact quote_version + acceptance evidence + payment gate reference	WON or blocked
Coordination	Orchestrator	Any agent task needing execution	Task contract + capability + context + policy	Route + specialist result
Orchestrator	Specialist	Eligible routed task	Execution envelope	Structured result
Specialist	Coordination	Result ready	Result contract + evidence + warnings/blocker	Validation + next transition

The Orchestrator remains the routing layer unless an explicit specialist-to-specialist path is part of the defined contract. Coordination remains responsible for workflow truth and handoff state.

25. Phase 1 Final Definition of Done

- [] A new configured-channel lead can be safely ingested, identified/deduplicated and represented as an authoritative CRM opportunity.
- [] Sales can qualify the lead and consolidate requirements with explicit confirmed facts, assumptions and unanswered questions.
- [] Scheduler can propose and book meetings against authoritative availability with idempotent verified provider state, reminders and evidence.
- [] Quotation Master can generate policy-compliant deterministic quotes from exact requirement versions, with immutable versioning, approval and delivery trace.
- [] Negotiation creates governed new versions for material changes; ambiguous acceptance is blocked.
- [] Exact quote-version acceptance is recorded with evidence and authority.
- [] Configured payment/approved-exception gate is verified before WON.
- [] Coordination provides durable workflow state, dependencies, retries, blockers, handoffs and recovery.
- [] Orchestrator routes by capability, permission, policy, health, cost and latency with auditable selection and safe fallback.
- [] WhatsApp/provider webhooks are authenticated, normalized and idempotent.
- [] Admin Panel provides Phase 1 operational visibility and controlled commands across the 34-screen blueprint, with backend authority and audit.
- [] Security, tenant isolation, RBAC, audit, provider failure, concurrency and negative paths are verified.
- [] Mandatory functional and E2E tests pass; no skipped critical test is counted as pass.
- [] The exact tested release candidate is the exact deployed build, with smoke evidence and rollback readiness.

At this point Phase 1 ends at the governed WON boundary. The Phase 2 Project Manager / downstream workforce may be attached only through a structured handoff; downstream Finance, Development, QA, Handover and Maintenance workflows are not considered active Phase 1 completion criteria.

Appendix A. High-Risk Business Rules That Must Never Be Bypassed

- [] Payment status is separate from quotation acceptance.
- [] Quote delivery is not quote acceptance.
- [] Result receipt is not task acceptance.
- [] Model/provider success is not business success.
- [] Time passing is not meeting completion.
- [] An approval applies to the exact governed action/version for which it was issued.
- [] Material scope/commercial changes create a new immutable version.
- [] Provider state is evidence, not sole AgencyOS identity.
- [] WhatsApp is communication evidence, not authoritative CRM truth.
- [] Admin UI actions are commands; backend is the authority.

Appendix B. Required Objective Evidence Package

Evidence	Minimum artifact
Repository audit	Versioned tracker + baseline commands + architecture map
Contracts	Schemas + contract tests + examples
Database	Migrations + schema/uniqueness/tenant tests
Workflow	State transition tests + event/task traces
Agents	Agent-specific DoD tests + integration traces
Integrations	Provider adapter tests + webhook auth/idempotency evidence
Admin Panel	Screen verification + permission/action evidence
Security	RBAC/tenant/secret/injection/rate-limit test results
E2E	F-01..F-15 pass evidence + failure-path evidence
Release	RC identifier + deployment smoke + rollback readiness